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CUBESAT DEVELOPMENT FOR CANYVAL-X MISSION

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CANYVAL-X mission is developed for demonstrating a virtual telescope alignment system. The main objective of this mission is to build an inertial alignment of a virtual telescope with respect to the Sun during a few minutes by using two CubeSats, Tom(2U) and Jerry(1U). Inertial alignment is to maintain relative position and relative attitude of the two satellites with respect to the target object. For achieving the inertial alignment, a vision alignment system, relative position control with thruster, relative orbit determination with inter-satellite link system and satellite separation system need to be developed. This study presents an overview of the mission concept and configuration systems of Tom and Jerry.

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I. Introduction

Current space telescopes have a single structure, and consequently, their focal length cannot be elongated sufficiently. Sometimes, this problem prevents the improvement of the resolution of space telescopes. In order to resolve this problem, the concept of Virtual Telescope is proposed. A Virtual Telescope consists of two spacecraft; one has a lens system, and another has a detector system. By using formation flying, they can be simplified as a virtual telescope system. Then, the relative orbit distance of two spacecraft can be a baseline of a virtual telescope system[1,2]. The most important point of Virtual Telescope is to build the inertial alignment with respect to a celestial object and to maintain it in space environment. Inertial alignment means that relative position and relative attitude of the two spacecraft are simultaneously aligned with a target.

Yonsei University's Astrodynamics and Control Laboratory (ACL) and NASA's Goddard Space Flight Center have collaborated in a study to resolve the limitation of existing space telescopes as mentioned above. For this research, they have developed and demonstrated the core technology of virtual telescope with the CANYVAL-X(CubeSat Astronomy by NASA and Yonsei using Virtual Telescope ALignment-eXperimet) mission.

II. Overview of CANYVAL-X Mission

The CANYVAL-X mission consists of two CubeSats, one of which serves as the lens satellite, and the other, as the detector satellite movement of a Virtual Telescope. The primary mission objective of CANYVAL-X is to build an inertial alignment system of virtual telescope with respect to the target object (the Sun) during a few minutes by using two CubeSats Tom (2U) and Jerry (1U). The secondary objective is to provide an opportunity for CubeSat development and flight operation by students. The objective of this mission is not to observe the Sun directly. This mission is to test the required technology for virtual telescope system in space environments. The result of experiment can be utilized for subsequent science demonstrations, CANYVAL[2] (CubeSat Astronomy by NASA and Yonsei using Virtual Telescope ALignment) and Solar X-ray Imager mission[3]. The two CubeSats for the CANYVAL-X mission will be launched in June 2016.

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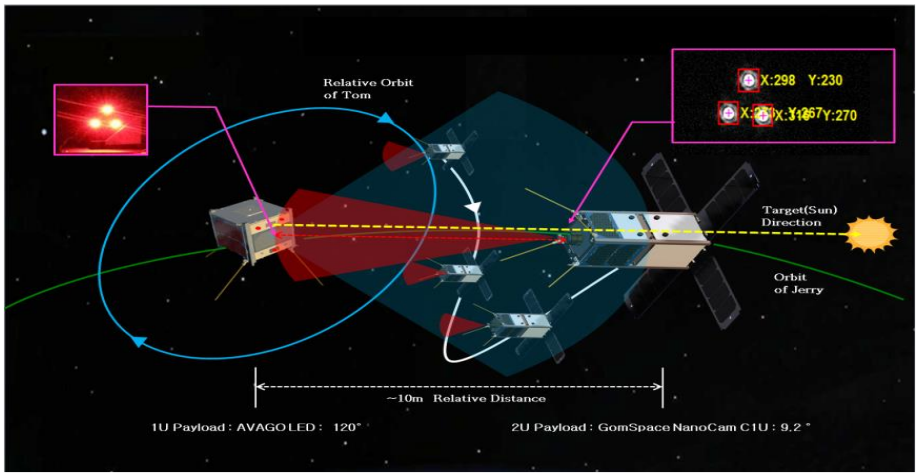


Figure 1. Concept of Inertial Alignment System of CANYVAL-X

III. Core Technologies for CANYVAL-X Mission

For constructing vision alignment in space, firstly, Tom needs the position information of Jerry in ECI (Earth Centered Inertial) frame. Next, Tom approaches Jerry and maintains a distance of 10m to 50m, which allows the operation of the vision alignment system. Finally, Tom and Jerry will make an inertial alignment by using relative orbit control. In order to minimize orbit control, two CubeSats will be launched out as one entity from a CubeSat deployer. After that, Tom and Jerry will be separated by a satellite separation device when the mission condition is satisfied. For these sequences, Tom and Jerry have to perform the following techniques; 1) Vision Alignment System, 2) Relative Orbit Determination System, 3) Relative Orbit Control System, 4) Satellite Separation System.

A. Vision Alignment System

This mission uses a vision alignment system that can determine the relative position and relative attitude between the two CubeSats at the same time. The vision alignment system consists of laser beacons on Jerry and one visual camera loaded on Tom. Tom detects the laser diode projection images to determine the relative position and relative attitude between the two CubeSats. In vision alignment mode, Jerry will fix its attitude for the laser diodes on Jerry such that they look at the Sun. Then, Tom will interrupt this line between Jerry and the Sun and will keep the inertial alignment between Jerry, Tom and the Sun during a few minutes. To do this, Tom will fix its attitude with respect to the Sun and will control its position on the transversal plane of Jerry and the Sun alignment by using reaction wheels and thrusters (Figure 1).

B. Relative Orbit Determination System

If Tom wants to approach Jerry, it needs to know the current position, velocity and time information of Jerry. The two CubeSats are each able to determinate absolute orbit by using GPS. After calculate the absolute orbit, Jerry sends the information to Tom via the UHF crosslink. Then, Tom receives this data along with the differential. This UHF communication is also used for telemetry and telecommanding between the ground station and CubeSats. For this purpose, main frequencies of two satellites were selected to have a difference of 0.02MHz. This difference can be identified by the ground station and also makes it possible to communicate with the same UHF Transceiver. (Figure 2).

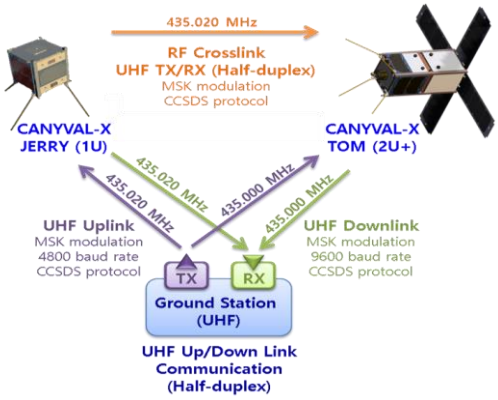


Figure 2. CANYVAL-X Communication System

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C. Relative Orbit Control System

To do proximity orbit control, Tom uses mCAT (micro Cathode Arc Thruster) which is a low thrust electric propulsion system developed by the George Washington University[4] and tested by NASA Goddard Space Flight Center (GSFC). The thruster system has four nozzles. The maximum impulse of each nozzle is 0.05 mNs. In order to obtain sufficient impulse, the four nozzles are located on same axis of the body frame. If Tom needs to orbit control, it first uses a three reaction wheel system for attitude control for thrust direction and then fires the proper thrust in each sampling time step. Figure 3 shows the configuration of the mCAT and the result of plasma firing in a vacuum chamber at 25 Hz.

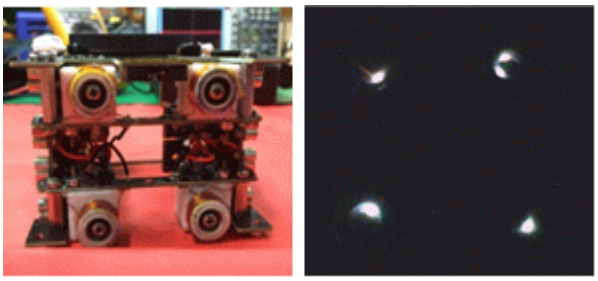


Figure 3. Configuration (left) of mCAT and Vacuum Testing (right)

D. Satellite Separation System

If Tom and Jerry are not one entity, they will gradually move away at a rate of 4.55 km/rev because of orbital perturbations. A separation device system is used to tie the two CubeSats. . Each satellite has a separation device mount on the z-axis of the body frame. Tom's mounting system consists of an electrical wire and heat resistor of 10 Ohm. The two satellites are bound by using dynamo wire, which carries high tension and is easy to melt with high temperature (Figure 4). Tom send a command to initiate at the switch at a voltage of 7.8V, then, the heat resistor is heated, melting the wire and disconnecting the system. Consequently, a kill switch spring of Jerry ejects Tom. This separation device is activated when the attitude of coupled satellite is oriented and aligned to a proper direction to minimize relative distance between satellites after separation.

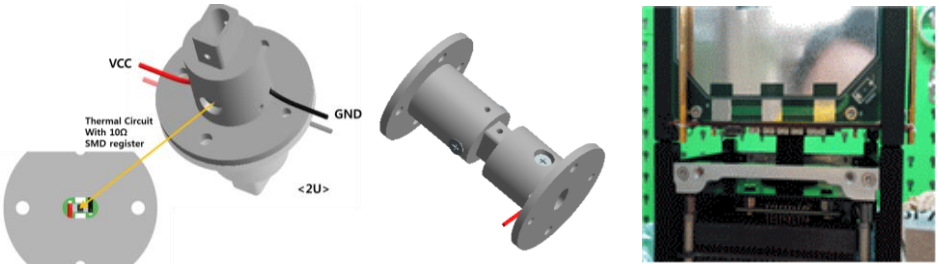


Figure 4. Separation device system

IV. Spacecraft System

The CANYVAL-X mission needs to develop two cube satellites for testing basic technologies of a virtual telescope in orbit. It has a limitation of resources for development. Consequently, the development philosophy of this mission follows proto-flight philosophy which results in developing one the proto-flight model. The CANYVAL-X system uses many COTS (Commercial Off-The-Shelf) products and several devices which are in-house, such as the solar panels, magnetic torquers, separation device, deployer devices, coarse sun sensors, a fine sun sensor and a thruster assembly. The mission life time is three months and both CubeSats need a three axis attitude control system and absolute orbit determination system. Flight softwares of Tom and Jerry are developed in free RTOS (Real Time Operation System) for embedded system programming that is simple and robust. The main data communication system is I2C (Inter-Integrated Circuit) via a

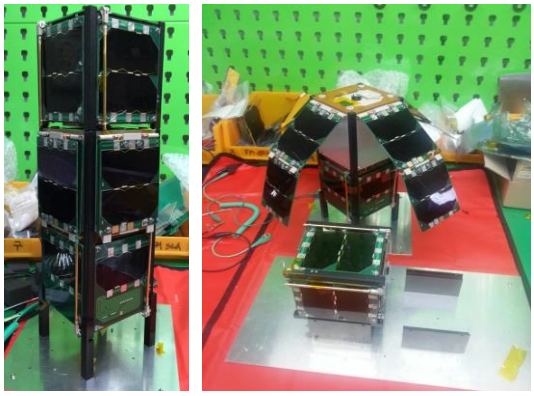


Figure 5. Coupled and Separated CubeSats

PC-104 connector. PWM, UART, and SPI are also used for sensor data connection and actuator control. Figure 6 shows a system interface diagram of the two CubeSats.

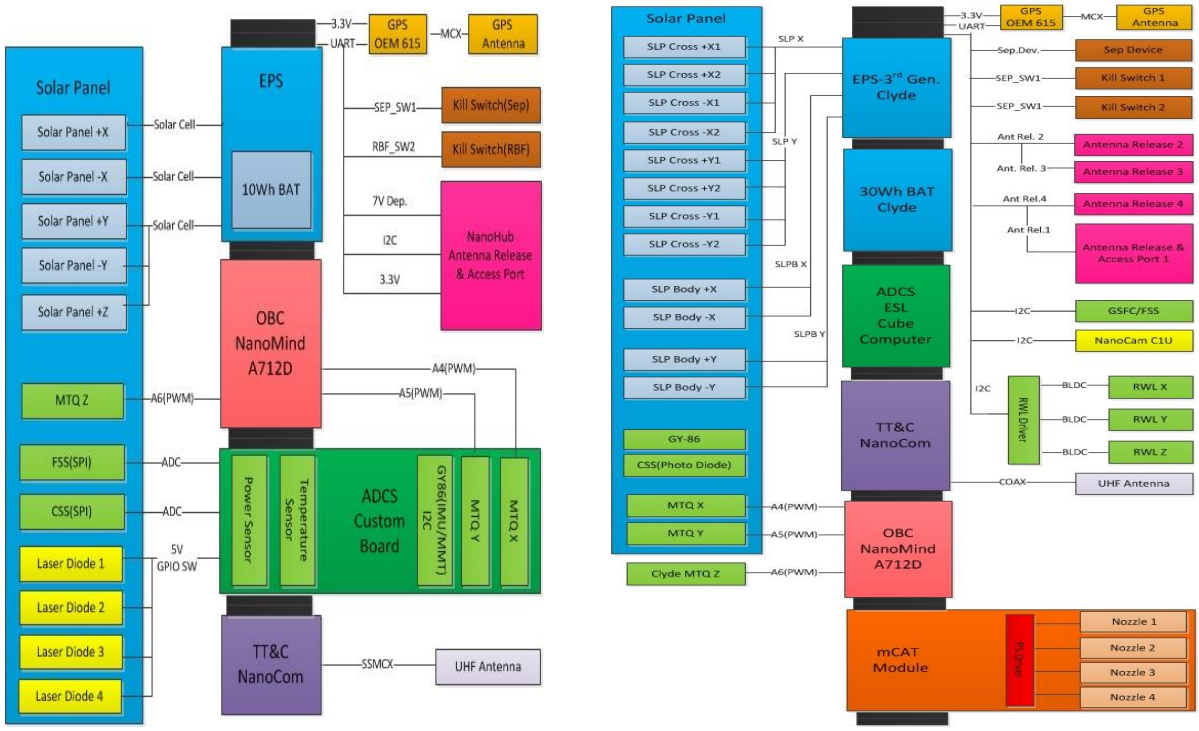


Figure 6. System interface diagram of Tom(right) and Jerry(left)

A. Jerry (1U)

The role of Jerry for this mission is to serve as a marker for vision alignment. The payload of Jerry consists of Laser LEDs that has a 120° FOV (Field of View). When in the vision alignment mode, Jerry turns on its LEDs and orients itself with respect to the Sun for the Vision Alignment System. For three axis attitude control, Jerry uses magnetic torquers for attitude control in each axis. The X- and Y-axis control are accomplished by using rod-type magnetic torquers, whereas the Z-axis is a PCB-type torquer. Four photo diodes serve as a coarse sun sensor and one fine sun sensor which are located on each solar panel for attitude determination. Jerry sends the position, velocity, and time information of the ECI frame to Tom every five minutes using Inter-Satellite links with UHF crosslink. Tom uses the data for differential orbit determination for a relative orbit determination. Table 1 shows the system characteristics of Jerry.

Table 1. System performance of Jerry

Properties	Value	
	Requirements	Performance
Mission Life Time	3 Month	
Payload	Laser Diodes	
Payload FOV	> 15.5°	120°(±60°)
Rel. Orbit Control	N/A	
Rel. Orbit Det.	N/A	
Attitude Control	< 5°	< 5°
Attitude Detetermination	< 5°	< 5°
Frequency	UHF	435.020 Hz
Uplink rate	4.8 kbps	4.8 kbps
Downlink rate	9.6 kbps	9.6 kbps
Link Margin	> 4 dB	> 5.51 dB
Protocol	CSP(CubeSat Protocolol)	
Mass	< 1.33 kg	0.81 kg
Power Generation	1.8W	2W
Deployment	Antenna	

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B. Tom (2U)

The payload of Tom consists of a visible camera for detecting the laser beacon provided by Jerry. It has a 610 nm to 660 nm band pass filter in order to eliminate ambient light. Tom mainly uses a three reaction wheel assembly for attitude control. It also has magnetic torquers for each of the three axes to dump angular momentum of wheels. For precise attitude determination when in vision alignment mode, it uses the fine sun sensor developed by NASA/GSFC. There are many mission procedures for Tom. In particular, mCAT, the reaction wheel system, GPS and camera are activated at the same time in ten minutes to perform an inertial alignment of virtual telescope with respect to the Sun. In order to accomplish this, Tom generates sufficient power with the cross type solar panel. To prevent interference with the UHF antenna, Tom's solar panels utilize a double deployable solar panel system. Table 2 shows the system characteristics of Tom.

Table 2. System performance of Tom

Properties	Value	
	Requirements	Performance
Mission Life Time	3 Month	
Payload	Visible Camera	
Payload FOV	> 7.5°	9.22°
Relative Orbit Control(20 m)	<1 m	<1 m
Relative Orbit Determination	±10 cm	< 10 cm
Attitude Control	< 1°	< 1°
Attitude Determination	< 1°	< 1°
Frequency	UHF	435.000 Hz
Uplink rate	4.8 kbps	4.8 kbps
Downlink rate	9.6 kbps	9.6 kbps
Link Margin	> 4 dB	> 5.51 dB
Protocol	CSP(CubeSat Protocol)	
Mass	< 2.66 kg	2.25 kg
Power Generation	2.2 W	2.5 W (Normal) 13 W (Sun Pointing)
Deployment	Antenna, Solar Panel	

V. Conclusion

This study presents the concept and system of the CANYVAL-X mission. The basic technologies necessary for a virtual telescope are demonstrated by Tom and Jerry. This study verifies that Tom and Jerry have satisfied the system requirements. The CANYVAL-X mission will serve as the baseline at the virtual telescope that is the next generation space telescope.

Acknowledgments

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